

Godot Agent — Writeup

Godot Game Generation Agent — Writeup

Problem statement

Turn a single natural-language prompt into a playable Godot 4.5.1 game, rendered live in a browser iframe. Two-column UI: prompt + streamed progress on the left, the running game on the right.

Prioritization

Four evaluation axes: prompt alignment, playability, aesthetic, and the user not feeling stuck. Ranking under a fixed time budget:

1. **Never feel stuck.** A minute of silence is fatal; a minute of visible per-step progress is fine. This forced SSE streaming as a hard requirement and set the design of the phase-card UI.
2. **Always ship something playable.** Never show a broken or empty iframe. This drove the fallback-first asset pipeline and the “budget-exhausted but shippable” synthesize exit.
3. **Aesthetic by default.** At least one shader, locked palette, style-sentence prefix on every asset prompt, juice list required.
4. **Alignment last** — the template picker and the GameDesign schema carry most of this, so it gets handled upstream.

Pipeline

Five phases, each emitting SSE events the frontend renders as live phase cards.

1. Design (~40–90 s). Opus 4.7 returns a `GameDesign` (Pydantic: dimension, template, controls, mechanics, camera, signals, palette, juice, assets, shaders, sfx_map). A Gemini Flash critic picks the winner from N candidates under an anti-blandness bias; fallbacks and sfx are validated against the bundled manifests so the model cannot hallucinate filenames. Current `TEMPERATURES = (0.7,)` — single candidate — because Anthropic’s 4K output-tokens-per-minute org cap is hit easily by best-of-3.

2. Assets (~30–90 s). `asyncio.gather` across every asset. Each task runs inside a generate-or-fallback wrapper. Timeouts: PixelLab 20 s (sprites), `gpt-image-1` 25 s (bg/UI), Tripo 60 s (meshes). Garbage detectors per service (near-uniform PNG, tiny GLB, bbox-diagonal-off-by-10×). A per-

session circuit breaker short-circuits a service to fallback after 2 consecutive failures. If both the API and the bundled Kenney asset are unavailable, a palette-themed PIL placeholder (vertical gradient for `bg`, solid mid-tone otherwise) is written; the game always ships.

3. Synthesize (~30–90 s). Template copied to the session workspace. Sonnet 4.6 receives the full file tree + `GameDesign` + resolved asset manifest + SFX manifest + API/shader snippets + per-template notes. Returns a strict JSON edit plan (`{files: [{path, content}]}`). Sanity pass checks: balanced braces, `extends` on scripts, `res://` refs resolve to real files (including `project.godot`), input actions declared, banned inline `load()` / `preload()` in `.tscn` / `.tres` (runtime parse error). Failures feed back into a repair loop.

4. Build (~15–30 s). `godot --headless --export-release "Web". threads=false` in the preset → single-threaded WASM, no COOP/COEP headers needed, runs in every browser.

5. QA (~15–25 s). Playwright Chromium polls `window.__GODOT_READY__` (set by the per-template `ready_signal autoload`), captures 3 screenshots at `ready+0/3/6s`, Gemini Flash reviews for blank-screen / missing-texture / broken-scale. Runs after the iframe is already visible; only triggers a repair on critical issues.

Global repair budget = 3 iterations total across synthesize + build + QA combined. Hard cap on worst-case latency and cost. `SessionManager` owns the pipeline as a detached `asyncio.Task`; SSE disconnect no longer cancels the job. A `/api/cancel/{sid}` endpoint gives the user explicit termination.

Load-bearing contracts

- **SSE event schema.** Every phase emits `{phase, step, status, detail, payload}`. This is the single contract the frontend, QA repair trigger, and history replay all key off — the phase cards, the detached pipeline, and `GET /api/stream/{sid}` resuming from a reconnect all fall out of one shape.
- **Workspaces as replayable history.** Every session persists `_log/design_candidates.json`, `_log/synthesize.json`, and `qa/*.png`. The full event stream can be re-served on reconnect (history replay before live events), and any past run is a static URL away. Makes debugging and the future gallery UI free.
- **Ready-signal autoload.** Every template ships an autoload that sets `window.__GODOT_READY__ = true` via JavaScriptBridge once the main scene loads. QA polls this instead of racing a timer — removes an entire class of flaky screenshot timing bugs.
- **Anti-blandness critic.** The Gemini Flash critic prompt is not a neutral rubric grader; it explicitly rejects hedging, generic verbs, empty juice, unpicked art direction. An opinionated taste filter is what keeps “make something fun” from collapsing into a generic asset dump.

- **Hot-reloadable prompts.** `prompts/*.md` are loaded per-call. Design and synthesize prompts can be tuned without a server restart, tightening the iteration loop.

How to run

```
./scripts/bootstrap.sh                                # Godot + export templates +  
    Playwright + uv sync  
cd backend && uv run uvicorn app:app --reload          # :8000  
cd frontend && npm run dev                             # :5173
```

Open `http://localhost:5173`, type a prompt, click Generate. Typical wall-clock: 2–3 min for 2D, 3–4 min for 3D. All sessions persist at `backend/workspaces/<sid>/web/index.html` and are static-served — bookmark to replay. Previously-promoted games can live in `backend/gallery/` (tracked in git, stripped of the 37 MB wasm).

Key decisions

- **Template-guided synthesis over from-scratch.** Forking polished community starters (Godot 2D/3D getting-started, GDQuest, Kenney kits) brings per-template craft from ~10 hrs to ~3. The template already runs; the LLM only customizes.
- **Strict schema + manifest-validated references.** `GameDesign` is a Pydantic model; every `fallback_role` must key into `fallback_assets/manifest.json` and every `sfx_map` value must key into `sfx_manifest.json`. The LLM cannot hallucinate a filename or a role that doesn't exist — validation failures are re-prompted with the violation.
- **Sanity check as primary quality gate.** Balanced braces, `extends` declarations, `res://` refs resolving to real files (including `project.godot`'s `main_scene`), banned `load()` / `preload()` in `.tscn` / `.tres`. This catches “exports cleanly, runs blank” bugs that Godot's own exporter will not.
- **Locked art_style + palette applied to every asset prompt.** Single style sentence and 4–6 hex codes are injected into every image/mesh prompt. Cheap to implement, single largest aesthetic-cohesion lever.
- **Shader snippets library.** Every game ships with ≥ 1 shader from a hand-tuned set (CRT, water, outline, dither, sky-gradient, palette-lock). Large aesthetic upgrade at near-zero per-game cost.
- **Generated-first, bundled-fallback assets.** The resolver always returns something; the pipeline cannot fail the asset phase. Combined with the repair budget, this produces a “best-effort but guaranteed-shippable” system.
- **Global repair budget (cap = 3).** Per-phase caps of 2 each stack to 6 worst-case; a single shared counter caps total wall-clock and API cost deterministically.
- **File-level generation + headless export**, not MCP / live editor. Faster, simpler, one-shot appropriate.

- **Single-threaded web export.** Eliminates SharedArrayBuffer + iframe CORS pain.
- **QA reads what the user sees.** Playwright on the web build, not Godot editor-mode — catches WebGL, font, and canvas-sizing bugs Godot itself doesn't.
- **Opus 4.7 for design, Sonnet 4.6 for synthesize, Gemini Flash for critic + QA.** Design needs taste + schema adherence; synthesize needs speed + 16k output tokens; QA needs cheap multimodal.
- **Streaming UX as a first-class feature.** Phase cards, per-asset thumbnails, live step details. The point isn't speed — it's that slow-but-progressing never feels stuck.

Bugs encountered during build

- **Renderer mismatch.** Templates shipped with `forward_plus` (requires WebGPU, broken on Chromium-web). Flipped all three tier-1 templates to `gl_compatibility`.
- **Blank canvas despite successful export.** `project.godot` referenced a `main_scene` whose file was never created. Sanity check extended to scan `project.godot` for `res://` refs.
- **Second blank canvas.** Scenes contained inline `load("res://sfx/foo.wav")` — valid in scripts, invalid in `.tscn / .tres` (runtime parse error). Sanity now bans `load()` / `preload()` inside scene/resource files.
- **Magenta placeholder assets.** When both API and bundled fallback fail, a PIL placeholder was drawn. Originally magenta + text label. Rewritten to a palette-themed gradient (bg) or mid-tone solid so fallback assets blend with the design's palette.
- **Cancel button broken.** After detaching the pipeline into a background task, SSE disconnect no longer terminated it. Added `POST /api/cancel/{sid}` + `CancelledError` handling in `SessionManager`.
- **Frontend phase cards stayed idle forever.** sse-starlette emits `\r\n\r\n` between events; the frontend was splitting on `\n\n`. Zero events were ever parsed. Split now uses `/\r?\n\r?\n/`. This was the single most misleading bug — backend was fully working; only the UI looked stuck.
- **PixelLab 422.** Designs specifying 24×24 pickups were rejected (API requires ≥32×32 area). Client now clamps width/height to ≥32.
- **Anthropic 429.** The synthesize repair loop fires up to 3 Sonnet calls back-to-back, tripping the 4K output-tokens-per-minute org cap and aborting the run. `claude_json` now retries up to 4× honoring `retry-after`, with exponential backoff.

Shortcomings + next steps

- **Single design candidate.** Best-of-3 is in the code but disabled to stay under the token-rate cap. With a higher cap or streamed token accounting, re-enabling it would measurably raise taste-floor on vague prompts.

- **3D is flakier than 2D.** Walker_3d has more scene surface (camera rig + physics + third-person controller), so Claude takes more repair rounds. Shrinking the template's wired surface or splitting synthesizer into two smaller calls (scene vs. scripts) would help.
- **No token-stream in the Design card.** Users see "fanout start" then silence for ~60 s. Streaming Claude chunks to the UI would remove the only remaining perceived-stuck window.
- **Gallery / session history is CLI-only.** A `GET /api/sessions` + sidebar list is ~30 min of work and would let users reopen past games without URL bookmarks.
- **Fallback asset packs ship empty.** Kenney manual download required per `bootstrap.sh`; only palette placeholders currently render. Non-blocking (pipeline ships regardless) but a polish hit.
- **No generated music.** Bundled loops only.
- **Pre-template coverage.** `shooter_2d`, `puzzle_2d`, `adventure_3d` exist in plan but aren't polished. Rhythm / card-game templates are absent; prompts in those genres will map to the closest tier-1 template and feel off.